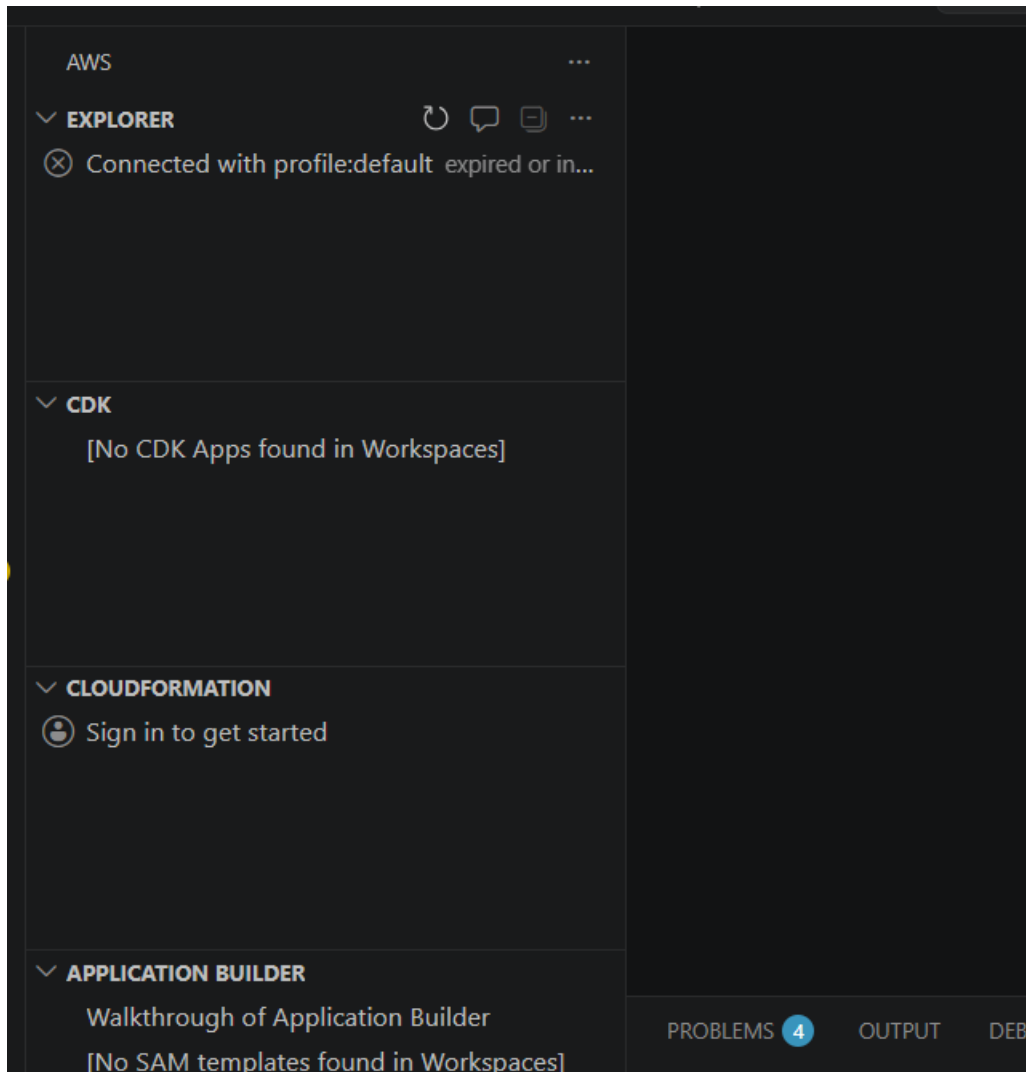
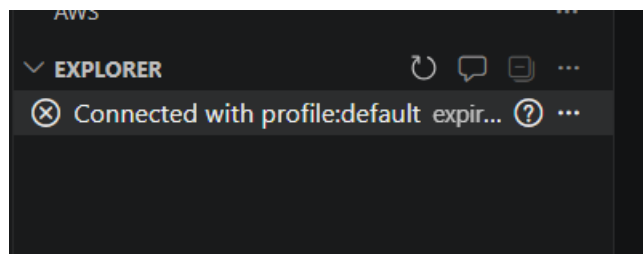


AWS Toolkit setup in Vscode

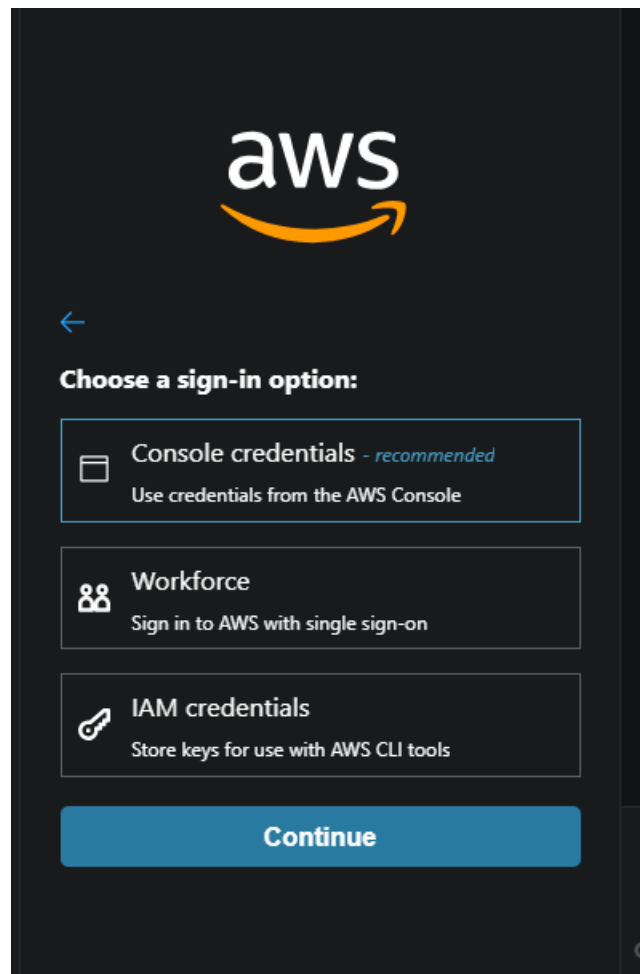
Install AWS Toolkit extension



Explorer => Add new connection



We will use IAM credentials



Go to AWS console ==> IAM ==> Create User group

)

Summary

banking-notification-group

June 01, 2026, 23:34 (UTC-04:00)

Permissions

Access Advisor

You can attach up to 10 managed policies.

Filter

 Search

All t

☐	Policy name ↗	Type
☐	  AmazonS3FullAccess	AWS managed
☐	  AmazonSNSFullAccess	AWS managed
☐	  AWSLambda_FullAccess	AWS managed
☐	  CloudWatchLogsFullAccess	AWS managed

Create IAM User and add to the banking-notification-group

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☒ **Add user to group**

Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ **Copy permissions**

Copy all group memberships, attached managed policies, and inline policies from an existing user.

☐ **Attach policies directly**

Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

User groups (1/1)

Q Search



Create group

< 1 > ⚙

☒	Group name ↗	Users	Attached policies ↗	Created
☒	banking-notification-group	0	CloudWatchLogsFullAccess , AmazonSNS...	2026-06-01 (2 minutes ago)

► Set permissions boundary - optional

Cancel

Previous

Next

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name
banking-notification-user

Console password type
None

Require password reset
No

Permissions summary

< 1 >

Name ↗	Type	Used as
banking-notification-group	Group	Permissions group

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel

Previous

Create user

Create User

Go to View user ==> Security credentials

Access keys ==> Create access key

Access keys (0)

Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. [Learn more](#)

No access keys. As a best practice, avoid using long-term credentials like access keys. Instead, use tools which provide short term credentials. [Learn more](#)

Create access key

Create access key

Enter Access key and Secret key in Vscode AWS Toolkit to finish the connection



←

IAM credentials:
Credentials will be added to the appropriate ~/.aws/ files
[Learn More](#)

Profile name
The identifier for these credentials
banking-notification-processor

Access key

Secret key

Access key best practices & alternatives [Info](#)

Avoid using long-term credentials like access keys to improve your security. Consider the following use cases and alternatives.

Use case

☒ **Command Line Interface (CLI)**

You plan to use this access key to enable the AWS CLI to access your AWS account.

☐ **Local code**

You plan to use this access key to enable application code in a local development environment to access your AWS account.

☐ **Application running on an AWS compute service**

You plan to use this access key to enable application code running on an AWS compute service like Amazon EC2, Amazon ECS, or AWS Lambda to access your AWS account.

☐ **Third-party service**

You plan to use this access key to enable access for a third-party application or service that monitors or manages your AWS resources.

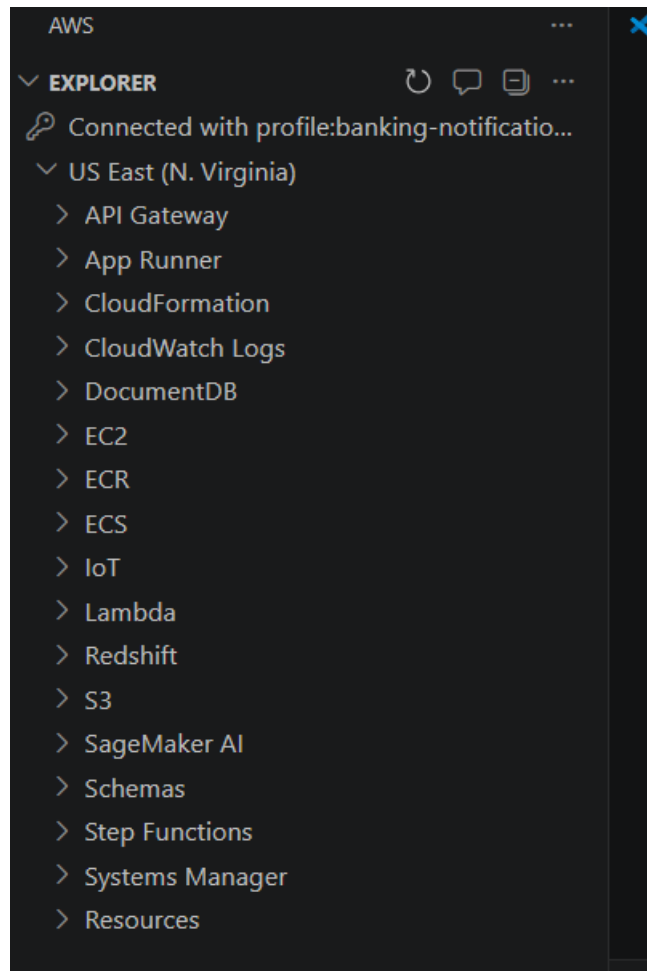
☐ **Application running outside AWS**

You plan to use this access key to authenticate workloads running in your data center or other infrastructure outside of AWS that needs to access your AWS resources.

☐ **Other**

Your use case is not listed here.

Now add connection ==> Add Access key and Secret key for the IAM user



Sign into Amazon Q

AMAZON Q

...

▼ NOTIFICATIONS (1)

...

🔗 Amazon Q Developer end of support

▼ LOGIN

...

no longer be available starting May 15, 2026,
but existing users can still sign-in below. [Learn
more](#)

Create New Account 🔒

existing users

Builder ID



Builder ID

Sign in with your existing Builder ID.



Identity Center

Sign in through your organization's IAM Identity
Center.

Sign in with existing account

I am using GitHub to connect



Sign in to **GitHub**
to continue to **Amazon Web Services (Builder ID)**

Username or email address

Password

[Forgot password?](#)

Sign in

or

 Continue with Google

 Continue with Apple

New to GitHub? [Create an account](#)

[Sign in with a passkey](#)

AWS Builder ID

Privacy & data

AWS takes your privacy seriously and the security of your information is our top priority. You can review more about how we handle your information in the [AWS Privacy Notice](#).

Request your data

Builder ID data

[Request your Builder ID data](#)

You can request your Builder ID profile information and the AWS application and services you have accessed by selecting Request your Builder ID data. We will process your request within 30 days, and notify you when your data is ready. While your data request is pending, you will not be able to delete your AWS Builder ID. [Learn more](#).

Deleting AWS Builder ID

[Delete AWS Builder ID](#)

When you delete your AWS Builder ID, you can no longer use your AWS Builder ID to access AWS applications that you previously accessed through AWS Builder ID. We will delete any remaining content associated solely with your AWS Builder ID and you will not be able to recover it afterwards.

I am not going to use Amazon Q to generate code. Next time will try
In case of Access Denied errors, add PowerUserAccess

Permissions policies (5)

Permissions are defined by policies attached to the user

☐	Policy name ↗
☐	  AmazonS3FullAccess
☐	  AmazonSNSFullAccess
☐	  AWSLambda_FullAccess
☐	  CloudWatchLogsFullAccess
☐	  PowerUserAccess

Create a new SNS topic on the AWS console:
BankingAlerts

SNS => Create Subscription

[Amazon SNS](#) > [Subscriptions](#) > Create subscription

New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#) [↗](#)

Create subscription

Details

Topic ARN

Protocol
The type of endpoint to subscribe

Endpoint
An email address that can receive notifications from Amazon SNS.

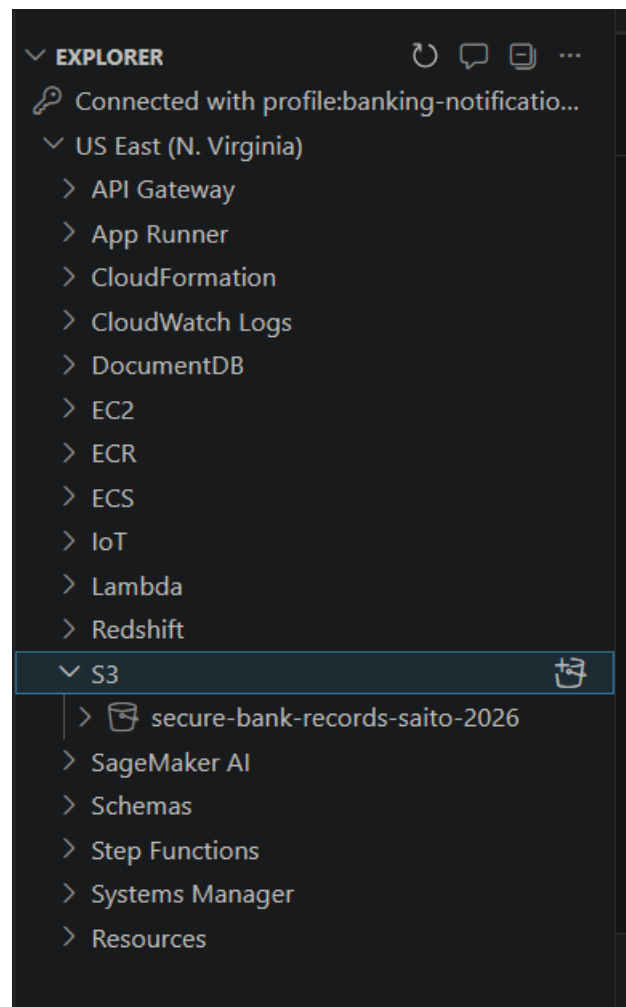
After your subscription is created, you must confirm it. [Info](#)

Subscription filter policy - optional [Info](#)
This policy filters the messages that a subscriber receives.

Redrive policy (dead-letter queue) - optional [Info](#)
Send undeliverable messages to a dead-letter queue.

[Cancel](#) [Create subscription](#)

Create S3 bucket



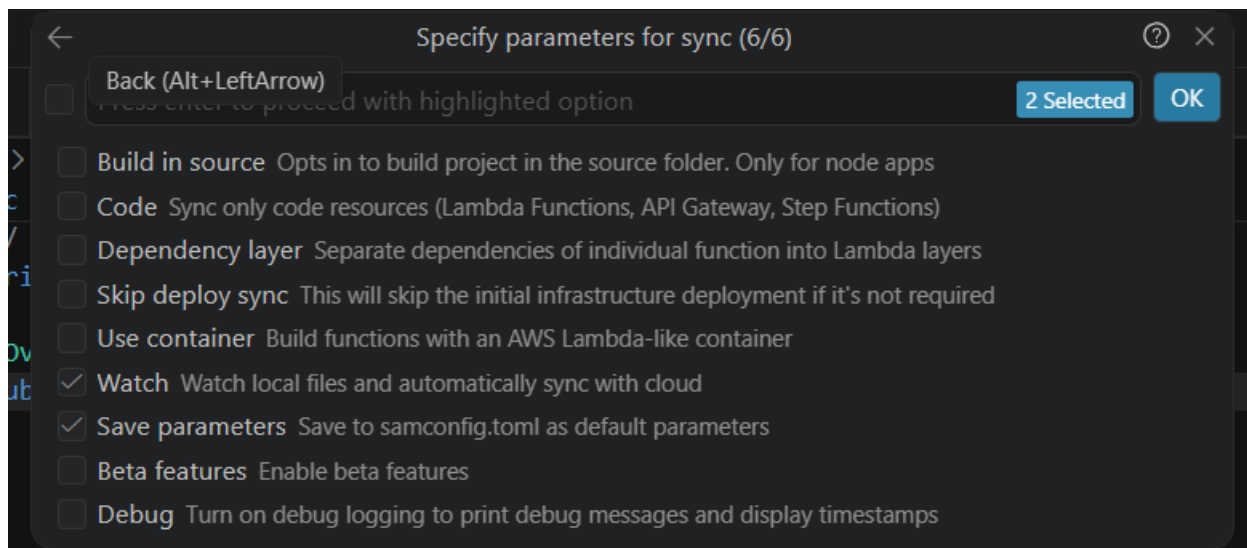
Install SAM CLI: https://www.google.com/url?sa=i&source=web&rct=j&url=https://github.com/aws/aws-sam-cli/releases/latest/download/AWS_SAM_CLI_64_PY3.msi&ved=2ahUKEwjNmZjE8-mUAXVig4kEHUizGrwQykOegYIAAgJEAE&opi=89978449&cd&psig=AOvVaw3iyQIl-iaL1QlibCKtr3A4&ust=1780535976207000

ctrl + shift + P

AWS: Deploy

us-east-1

then enter a Stack name: banking-notification-stack



Click Ok

in case of error: `sam deploy --template-file bank-processor/template.yaml --guided`

```
PS C:\Users\saito\source\repos\Telusko_Java_Enterprise_Spring\JavaEE_SpringMVC_Microservices\AWS toolkit> sam deploy --template-file bank-processor/template.yaml --guided

Configuring SAM deploy
=====

Looking for config file [samconfig.toml] : Found
Reading default arguments : Success

Setting default arguments for 'sam deploy'
=====
Stack Name [banking-notification-stack]:
AWS Region [us-east-1]:
#Shows you resources changes to be deployed and require a 'Y' to initiate deploy
Confirm changes before deploy [y/N]: Y
#SAM needs permission to be able to create roles to connect to the resources in your template
Allow SAM CLI IAM role creation [Y/n]: Y
#Preserves the state of previously provisioned resources when an operation fails
Disable rollback [y/N]: Y
```

SAM CLI is bit of a hassle, I am going to create Lambda function on AWS Console itself

Create function [Info](#)

Choose one of the following options to create your function.

☒ Author from scratch

Start with a simple Hello World example.

☐ Use a blueprint

Build a Lambda application from sample code and configuration presets for common use cases.

☐ Container image

Select a container image

Basic information

Function name

Enter a name that describes the purpose of your function.

BankingNotificationProcessor

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime [Info](#)

Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Java 17



Permissions

By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. To use a different role, choose **Custom execution role** under **Additional settings**.

Custom settings [Info](#)

Durable execution - [new](#)

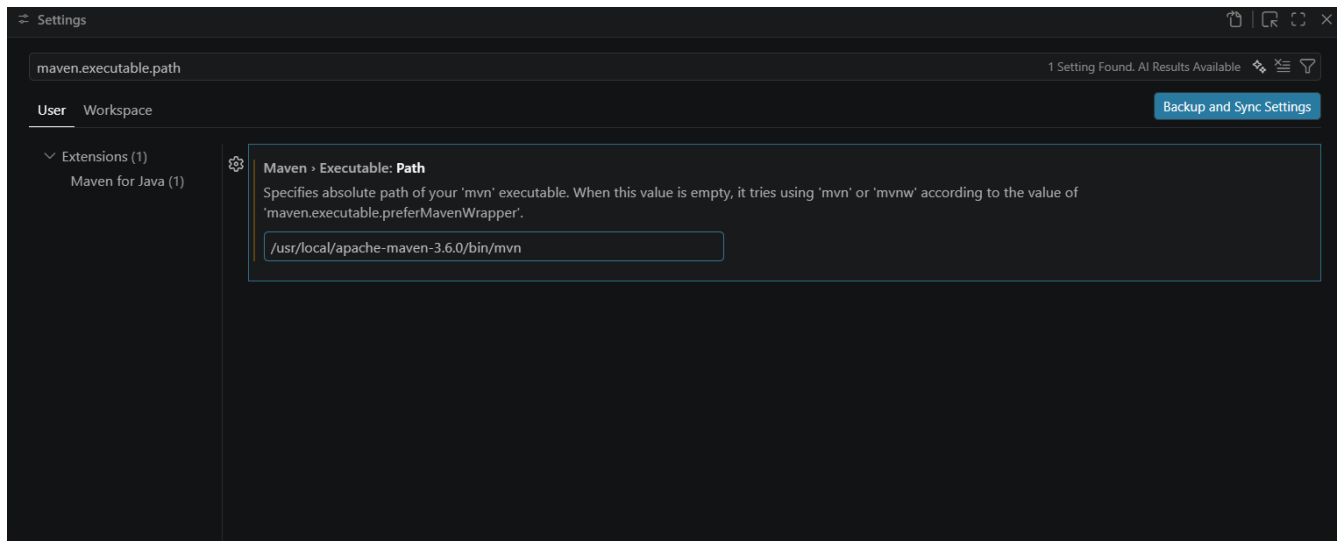


Build resilient, stateful applications with automatic failure recovery.

EC2 capacity provider - [new](#)

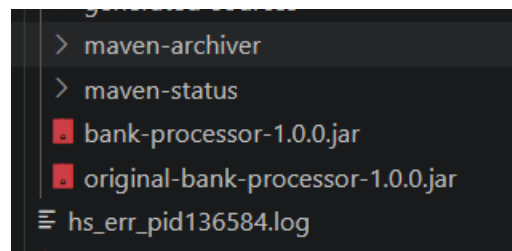
Run Lambda on your preferred EC2 instance types instead of Lamb

► Additional settings

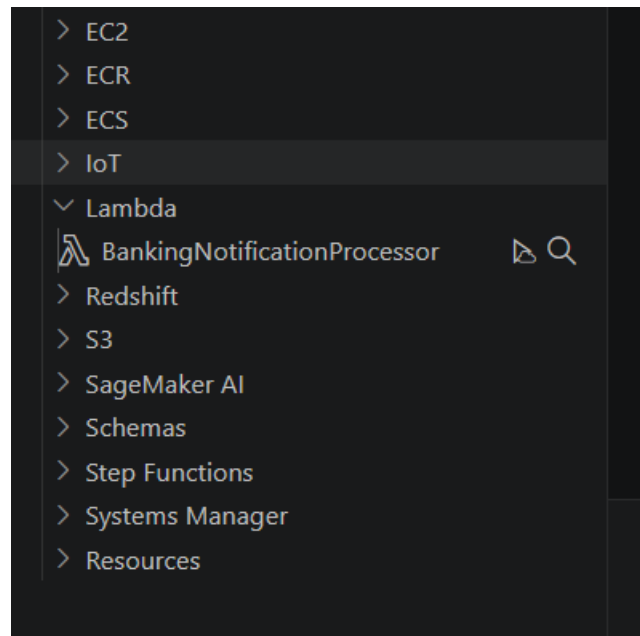


mvn clean package "-Dmaven.test.skip=true"

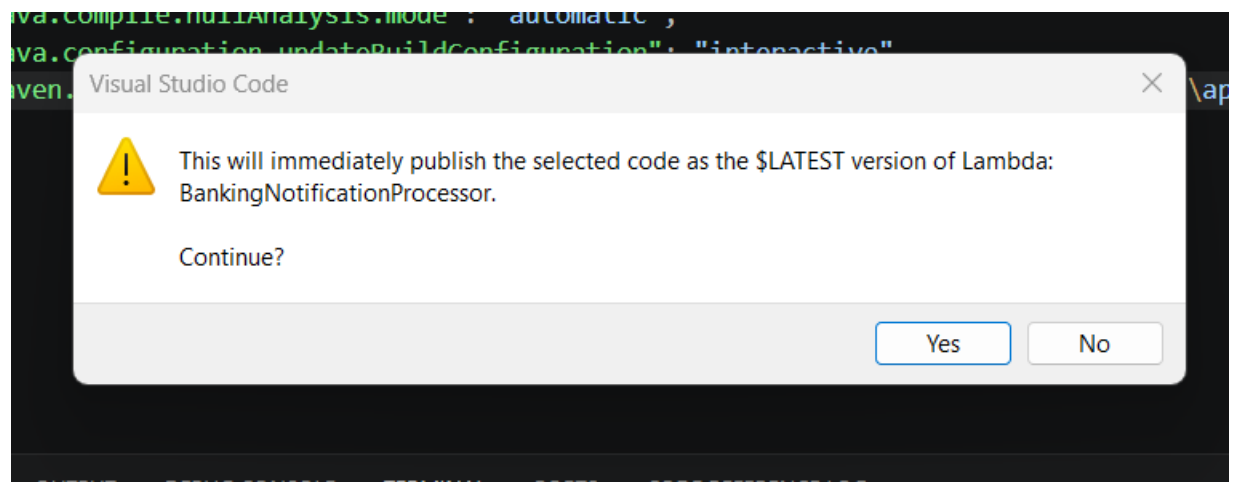
You will see bank-processor-1.0.0.jar



Now I see Lambda here



Right-click Lambda function and upload Lambda



☰ Invoke Lambda BankingNotificationProcessor

Invoke Lambda BankingNotificationProcessor

Remote invoke configuration

Remote Invoke

Function Name: BankingNotificationProcessor

Resource ARN

arn:aws:lambda:us-east-1:577638386543:function:BankingN

Region

us-east-1

Runtime

java17

Remote debugging

☐ Remote debugging is not recommended for production environments. The AWS function by deploying it with an additional layer to enable remote debugging. You breakpoints are then used to step through the remote function invocation. [Learn](#)

Payload

Enter the JSON payload for your Lambda function invocation. You can **Load sample e** templates, **Load local file** from your computer

Runtime settings [Info](#)

Runtime
Java 17

Handler [Info](#)
example.Hello::handleRequest

Architecture [Info](#)
x86_64

[Edit](#)

[Edit runtime management configuration](#)

▼ Runtime management configuration

Runtime version ARN [Info](#)
[arn:aws:lambda:us-east-1:runtime:9357d9b95184e01dcf0a5539c1454d9ff5f61c23f32908019e94f6eb78162e96](#)

Update runtime version [Info](#)
Auto

Layers (0) [Info](#)

Last fetched 6/2/2026, 9:49:13 PM [Refresh](#) [Edit](#)

A new runtime is available for your function's language: Java 25

Handler [Info](#)

com.awstoolkit.TransactionHandler::handleRequest

You can change either the function's runtime or the instruction set architecture in one update. To update both, you must repea

Architecture [Info](#)

Choose the instruction set architecture you want for your function code.

Note: another issue, ZIP file cannot have .Jar file instead we need .class

```
jar xf ..\bank-processor-1.0.0.jar
```

```
Remove-Item target\bank-processor-lambda.zip -ErrorAction SilentlyContinue
```

```
Copy-Item target\bank-processor-1.0.0.jar target\bank-processor-lambda.zip
```

I reupload into Lambda

I create inLine policy for the Lambda role to give access to S3

[BankingNotificationProcessor-role-23pu1wca](#) > Create policy

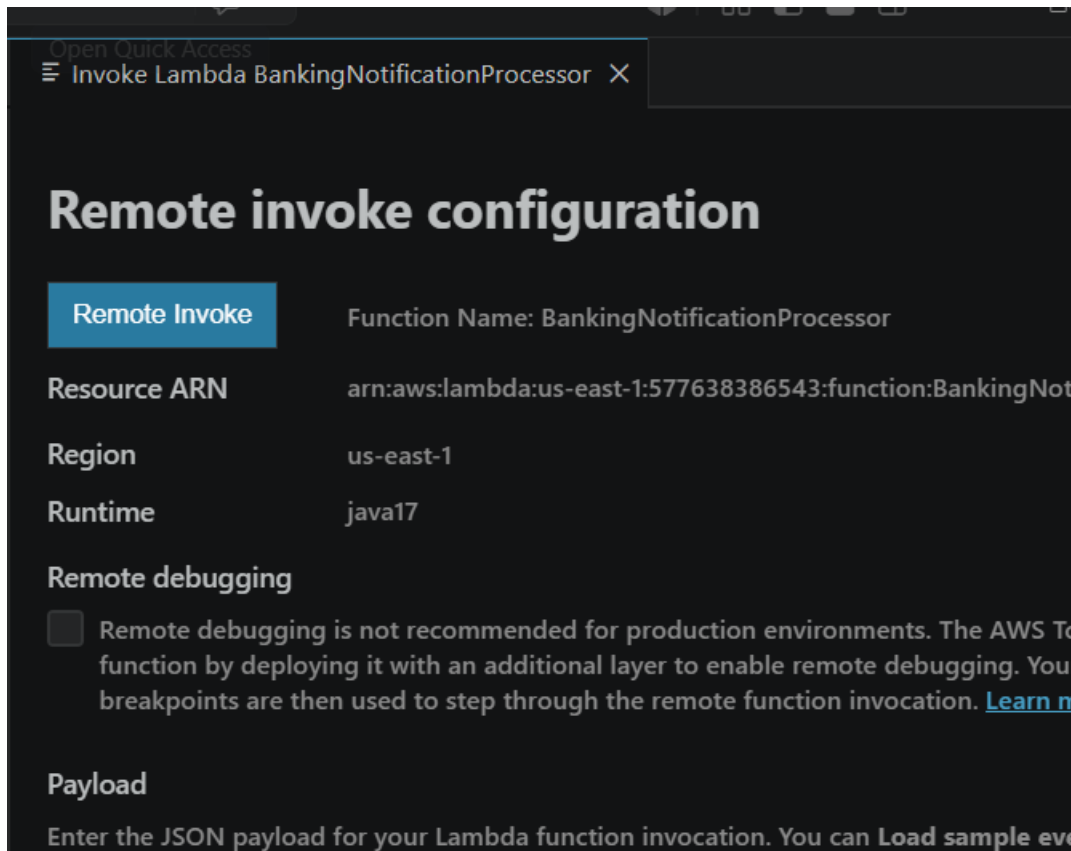
ns

Specify permissions [Info](#)

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using

Policy editor

```
1  {
2    "Version": "2012-10-17",
3    "Statement": [
4      {
5        "Sid": "AllowReadTransactionFilesFromS3",
6        "Effect": "Allow",
7        "Action": [
8          "s3:GetObject"
9        ],
10       "Resource": "arn:aws:s3:::secure-bank-records-saito-2026/*"
11     },
12     {
13       "Sid": "AllowListTransactionBucket",
14       "Effect": "Allow",
15       "Action": [
16         "s3:ListBucket"
17       ],
18       "Resource": "arn:aws:s3:::secure-bank-records-saito-2026"
19     }
20   ]
21 }
```

Test Payload for Lamda

```
{
  "Records": [
    {
      "eventVersion": "2.1",
      "eventSource": "aws:s3",
      "awsRegion": "us-east-1",
      "eventTime": "2026-06-02T23:55:00.000Z",
      "eventName": "ObjectCreated:Put",
      "s3": {
        "bucket": {
          "name": "secure-bank-records-saito-2026",
          "arn": "arn:aws:s3:::secure-bank-records-saito-2026"
        },
        "object": {
          "key": "transaction_001.json",
          "size": 1024
        }
      }
    }
  ]
}
```

SNS Inline policy for Lambda role

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowReadTransactionFilesFromS3",
      "Effect": "Allow",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::secure-bank-records-saito-2026/*"
    },
    {
      "Sid": "AllowListTransactionBucket",
      "Effect": "Allow",
      "Action": "s3:ListBucket",
      "Resource": "arn:aws:s3:::secure-bank-records-saito-2026"
    },
    {
      "Sid": "AllowPublishBankingAlerts",
      "Effect": "Allow",
      "Action": "sns:Publish",
      "Resource": "arn:aws:sns:us-east-1:577638386543:BankingAlerts"
    }
  ]
}
```

[Code](#) | [Test](#) | [Monitor](#) | [Configuration](#) | [Aliases](#) | [Versions](#)

 Executing function: succeeded ([logs](#))

▼ Details

"Transaction Processed Successfully"

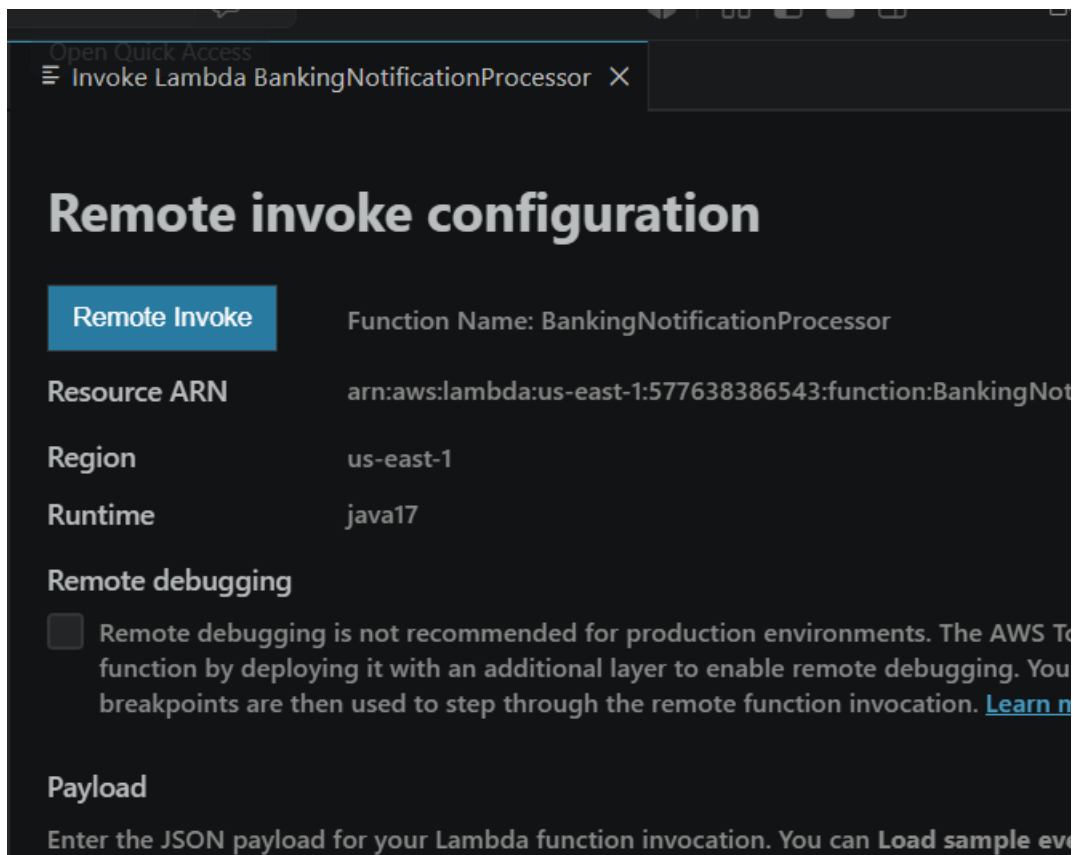
Summary

Code SHA-256 bvKjxqy0E548fkMY/wY6OssCNOMw1WFjDug9ni9Aumc=	Execution time 21 seconds ago
Function version \$LATEST	Request ID 469dd66f-12c2-408d-8f98-e1760484aee0
Duration 279.10 ms	Billed duration 280 ms
Resources configured 512 MB	Max memory used 166 MB

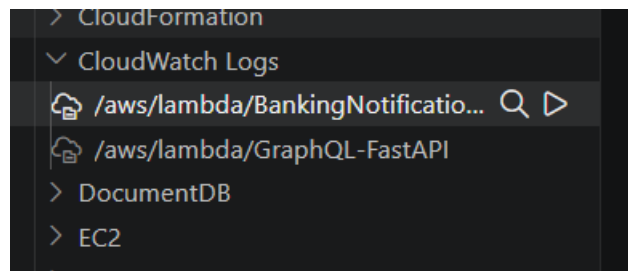
Log output

The area below shows the last 4 KB of the execution log. [Click here](#) to view the corresponding CloudWatch log group.

```
START RequestId: 469dd66f-12c2-408d-8f98-e1760484aee0 Version: $LATEST
[CloudWatch INFO] Lambda execution triggered by S3 Event.
[CloudWatch INFO] Attempting to process file: transaction_001.json from bucket: secure-bank-records-saito-2026
```



Each time I invoke from VSCode, CloudWatch generates a report



Add S3 Trigger in Lambda

✓ The test event "TestEvent" was successfully saved.
Notifications 🔊 0 🔊 0 🔊 5 🔊 0 🔊 0 ⌵

Trigger configuration [Info](#)

S3

aws asynchronous storage

Bucket
Choose or enter the ARN of an S3 bucket that serves as the event source. The bucket must be in the same region as the function.

✕
🔄

Bucket region: us-east-1

Event types
Select the events that you want to have trigger the Lambda function. You can optionally set up a prefix or suffix for an event. However, for each bucket, individual events cannot have multiple configurations with overlapping prefixes or suffixes

PUT ✕

Prefix - optional
Enter a single optional prefix to limit the notifications to objects with keys that start with matching characters. Any [special characters](#) must be URL encoded.
e.g. images/

Suffix - optional

What's the point of AWS Toolkit if I have to do everything in AWS Console

Monitor | Configuration | Aliases | Versions

Triggers (1) [Info](#)

☐

Trigger

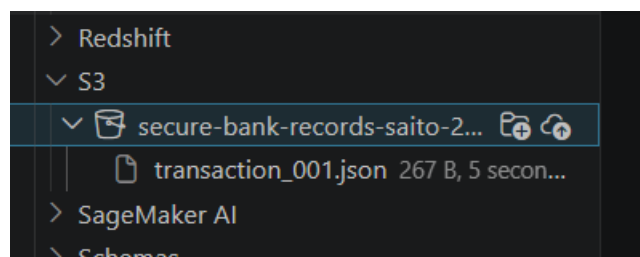
☐

S3: [secure-bank-records-saito-2026](#)

arn:aws:s3:::secure-bank-records-saito-2026

▶ Details

I uploaded new file into S3 bucket

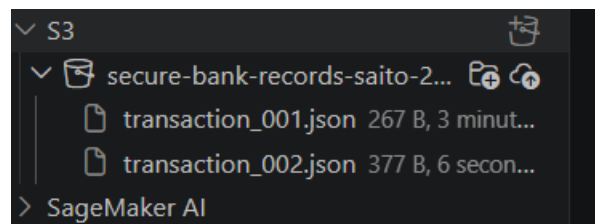


It has triggered Lambda automatically.

Check CloudWatch logs

```
2026-06-03T04:36:03.371-04:00 SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
2026-06-03T04:36:04.642-04:00 START RequestId: edbcc426-37f0-419d-9703-a317f18dcc62 Version: $LATEST
2026-06-03T04:36:04.838-04:00 [CLOUDWATCH INFO] Lambda execution triggered by S3 Event.
2026-06-03T04:36:04.877-04:00 [CLOUDWATCH INFO] Attempting to process file: transaction_001.json from bucket: sec
2026-06-03T04:36:07.543-04:00 [CLOUDWATCH SUCCESS] Successfully downloaded file payload from S3:
{
  "transactionId": "TXN001",
  "customerId": "CUST1001",
  "accountNumber": "1234567890",
  "amount": 250.75,
  "currency": "CAD",
  "transactionType": "DEBIT",
  "merchant": "Amazon Canada",
  "status": "APPROVED",
  "timestamp": "2026-06-02T23:55:00Z"
}
2026-06-03T04:36:08.420-04:00 [CLOUDWATCH SUCCESS] SNS Alert successfully published for: transaction_001.json
2026-06-03T04:36:08.439-04:00 END RequestId: edbcc426-37f0-419d-9703-a317f18dcc62
2026-06-03T04:36:08.439-04:00 REPORT RequestId: edbcc426-37f0-419d-9703-a317f18dcc62 Duration: 3797.02 ms    Bill
Load newer events...
```

I add the second file



```
2026-06-03T04:38:01.231-04:00 [CLOUDWATCH INFO] Attempting to process file: transaction_002.json from buc
2026-06-03T04:38:01.363-04:00 [CLOUDWATCH SUCCESS] Successfully downloaded file payload from S3:
{
  "transactionId": "TXN999",
  "customerId": "CUST9001",
  "accountNumber": "9876543210",
  "amount": 9500.00,
  "currency": "CAD",
  "transactionType": "WIRE_TRANSFER",
  "merchant": "Unknown Offshore Account",
  "status": "REJECTED",
  "riskScore": 98,
  "reason": "Suspicious high-value transfer to unknown destination",
  "timestamp": "2026-06-02T23:58:00Z"
}
2026-06-03T04:38:01.517-04:00 [CLOUDWATCH SUCCESS] SNS Alert successfully published for: transaction_002.
2026-06-03T04:38:01.520-04:00 END RequestId: 6eb4d644-4717-4013-97d2-3710d1f5728d
```

SNS triggered automatic emails

AWS Notification Message Inbox x



AWS Notifications <no-reply@sns.amazonaws.com>

to me ▾

Bank Notification: File [transaction_001.json] was successfully compiled and stored.

--

If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe:

<https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:577638386543:BankingAlerts:adc83a60-d168-4c54-aef9-a99fff4f948d&Endpoint=sai.toronto8@gmail.com>

Please do not reply directly to this email. If you have any questions or comments regarding this email, please contact us at <https://aws.amazon.com/support>



AWS Notifications <no-reply@sns.amazonaws.com>

to me ▾

Bank Notification: File [transaction_002.json] was successfully compiled and stored.

...

--

If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe:

<https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:577638386543:BankingAlerts:adc83a60-d168-4c54-aef9-a99fff4f948d&Endpoint=sai.toronto8@gmail.com>

Please do not reply directly to this email. If you have any questions or comments regarding this email, please contact us at <https://aws.amazon.com/support>
